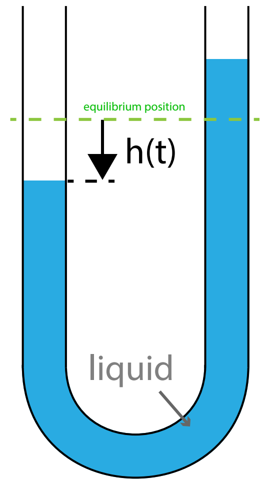
a manometer, used to measure pressure, is a U-shaped tube with liquid trapped inside. both ends of the tube are open.

suppose the right-hand side of the tube is always exposed to the atmosphere. as for the left side, it was exposed to a pressurized gas line for $t < 0$. but, at $t = 0$, the line to the high-pressure gas was disconnected. thus, the left side is exposed to the atmosphere as well for $t \geq 0$. consequently, the liquid in the tube at $t = 0$ initializes (a) at rest and (b) with a taller height on the right side.

our goal is to model $h = h(t)$ [m], the liquid level with respect to the equilibrium position. by convention, $h > 0$ implies the liquid level is lower on the left side.



1. derive a dynamic model for $h = h(t)$ [m] for $t \geq 0$ by writing a force balance (mass $\times$ acceleration = sum of forces) on the liquid. account for (i) the gravitational force and (ii) the viscous forces internal to the liquid. the [constant] parameters are listed in the table below. consider

   - the mass of liquid in the tube, which has a circular cross-section, is $\rho\pi R^2 L$ [kg].
   - the restoring force from gravity, due to displaced liquid, is $\pi R^2 \rho g 2h(t)$ [N].
   - the viscous force, which slows acceleration is $8\pi\mu L \frac{dh}{dt}$ [N] (the Hagen-Poiseuille equation).

   you should arrive at a second-order ordinary differential equation (ODE) for $h(t)$.

parameter	description
	manometer setup
R	radius of the tube [m]
L	total length of liquid in tube [m]
	properties of the liquid
ρ	density [kg/m^3]
μ	viscosity [N·s/m^2]
	a physical constant
g	standard acceleration due to gravity [m/s^2]

2. write the second-order ODE for $h(t)$ as a system of coupled, first-order ODEs in matrix-vector form:

$$\frac{d\mathbf{x}(t)}{dt} = A\mathbf{x}(t).$$

   the coefficient matrix A should be written in terms of the parameters R, μ, ρ, etc.

3. suppose the manometer tube (on earth, g=9.8 m/s^2) has $R = 0.01$ m and $L = 0.2$ m; olive oil ($\rho = 997$ kg/m^3, $\mu = 0.001$ Pa·s) inside; and begins with $h_0 = 0.05$ m. compute the eigenvalues and eigenvectors of the coefficient matrix A in Julia (they should be complex). based on these eigenvalues and eigenvectors, write the solution to $\mathbf{x}(t)$ in the time domain. plot $h(t)$ in Julia. for writing out $h(t)$, you may use only one significant figure from the eigenvalues/vectors.